Assessment: Student Instructions

Web Development Assessment: Backend API without a Database

1. What you will do

You get one question paper and build a REST API for it from scratch, in **Express** or **Spring Boot**. When it is ready, you run a small program called the **evaluator** on your own laptop. It runs 100 automated checks on your API and sends the result to your trainer.

The evaluator shows each check as PASS or FAIL, but no marks. When it finishes, it tells you your submission is recorded. Your trainer sees the result.

2. Before the day

1. Install **Node.js 18 or newer**, even if you use Spring Boot. Check with `node --version`.
2. Make sure your editor, Express or Maven and JDK, and Postman (or `curl`) work.
3. Practise P1 and P2 until you can build an API without looking anything up.
4. Bring your own laptop, fully charged, with its charger.

3. On the day

1. The invigilator gives you a **question paper** and a **PIN slip**. Your paper decides your set.
2. Join the classroom WiFi and stay connected until you finish.
3. Extract the **assessment kit** zip into a new folder. No `npm install` needed.
4. In the kit folder run `npm run whoami`. Enter your **registration number**, your **6-digit PIN** and the **set on your paper** (for example A5). It replies `Your set is A5`.
5. Create your API project in a **separate folder**, not inside the kit.

Your PIN is personal and works on **one laptop only**. Do not share it. If someone else uses your PIN, your trainer will see it and both of you may get zero. If you typed something wrong, run `npm run whoami -- --change`.

4. Rules

1. Build from scratch. No starter code, no copied code.
2. No database, no files, no seed data. Records live in memory and the server starts **empty**.
3. No frontend.
4. Port **8080** (read P0RT, default 8080).
5. Use exactly the paths, field names and status codes in your paper.
6. **Short on time?** Build POST, GET, PUT, PATCH and DELETE first. For pagination, filters, search and sort, at least return the first 10 records as shown in section 5 of your paper.
7. Do not edit, copy or share any file in the kit.
8. Work alone and follow your invigilator's instructions.

5. Running the evaluator

1. Restart your API so it is running on port 8080 and **empty**.
2. In the kit folder run `npm run evaluate`.
3. Wait up to about three minutes. You will see a counter like `Running checks: 37 of 100 attempted`, then a list of every check with PASS or FAIL. Do not stop your server.
4. **Restart your server before every run.**

Other port? Use `npm run evaluate -- --url http://localhost:9000`.

6. What the messages mean

Message	What to do
`Submission recorded. Your result has been sent to your trainer.`	Done. This is the normal ending.
`Cannot reach your server`	Nothing was submitted. Start your server on the right port and run again.
`Your server is not empty...`	Nothing was submitted. Restart your server and run again.
`Your result is saved on this computer but NOT recorded yet`	Check your WiFi, tell the invigilator, then run `npm run submit-pending`. Keep the kit folder as it is.
`This PIN is already registered to another laptop`	Tell the invigilator.

7. Good to know

- Every run is recorded. Your trainer decides which one counts, so run when your API is complete.
- The evaluator uses random data, so hard-coding answers will not work.
- If your API crashes during a run, the remaining checks fail. Bad input must never crash your server.
- Your trainer may re-run your code and ask you to explain it.
- A FAIL tells you which check failed, not why. Read that part of your paper again and test it with Postman.

8. Suggested plan

1. Read the whole paper (sections 3 to 6 are the contract, section 7 has examples).
2. Build POST and GET by id, then the list, PUT, PATCH and DELETE. Test each by hand.
3. Add pagination, filters, search and sort.
4. Check the “rules that are easy to miss” list in your paper.
5. Restart, run the evaluator, wait for “Submission recorded”.
6. Keep time for a second run if the first could not be submitted.

9. Quick reference

Task	Command
Log in / check your set	`npm run whoami`
Correct your details	`npm run whoami -- --change`
Run the evaluator	`npm run evaluate`
Another port	`npm run evaluate -- --url http://localhost:9000`
Send a saved result	`npm run submit-pending`
Start Express / Spring Boot	`node index.js` / `mvn spring-boot:run`